

Sedimentology Field Report 2024



ES3202

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Acknowledgements :

This Sedimentology Field Report for the year 2024, conducted under the auspices of the Department of Earth Sciences at IISER Kolkata, has been an enlightening journey. It is with immense gratitude that I extend my thanks to all those who have contributed to the success of this endeavor.

First and foremost, I would like to express my sincere appreciation to our esteemed instructors, Prof. Sanjay Kumar Mandal. Their expertise, guidance, and unwavering support throughout the field trip have been invaluable. Their dedication to teaching and mentorship have made this experience both enriching and memorable.

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Furthermore, I extend my heartfelt thanks to all my fellow group members, batchmates & Their collaboration, camaraderie, and shared enthusiasm have made this journey enjoyable and productive. Together, we have navigated the complexities of sedimentological analysis and emerged with a deeper understanding of the subject.

Lastly, I would like to acknowledge the Department of Earth Sciences at IISER Kolkata for providing us with the opportunity to undertake this field trip. Their commitment to fostering academic excellence and experiential learning has been instrumental in shaping our understanding of sedimentology.

In conclusion, I am grateful for the invaluable lessons learned and experiences gained during this sedimentology field trip. It has been a privilege to be a part of this journey, and I look forward to applying the knowledge and skills acquired here in my future studies and endeavors.

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CHAPTER 1. Introduction

Sedimentology is the study of rocks made from sediments and how they're formed by natural processes. It helps us understand how the Earth's surface has changed over time. This Sedimentology Lab Report is about what we learned during a trip to the Mohand River basin in the Siwalik Himalayas. under the guidance of our instructors, provided us with a hands-on opportunity to explore the geological complexities of this region.

Situated within the vast expanse of the Himalayan orogeny, the Siwalik basin offers a unique geological setting for studying sedimentary activities. Originating from the collision of the Eurasian and Indian plates, the rise of the Himalayas gave birth to the foreland basin, where fluvial systems transported sediments from the hinter Himalayan region to form the Siwalik range.

During our field trip, we focused on the Mohand River basin, a tributary of the Yamuna River, which spans the Lower to Middle Siwalik regions. Through meticulous observation and analysis, we examined sedimentary structures and depositional environments, gaining insights into the geological evolution of the area.

Among all these divisions, the Siwalik was once part of the great Indian plain. The fluvial system coming from the ranges of the Himalayas brought the sediments from the hinter Himalayan part and deposited them in today's Siwalik range. Furthermore, the Siwalik started to rise due to the Himalayan Frontal Thrust (HFT). The HFT created pressure on the deposited sediments, and the folding generated within them eventually formed the Siwalik. These Siwalik ranges can be grouped into three sub-groups:

1. The Upper Siwalik
2. The Middle Siwalik
3. The Lower Siwalik

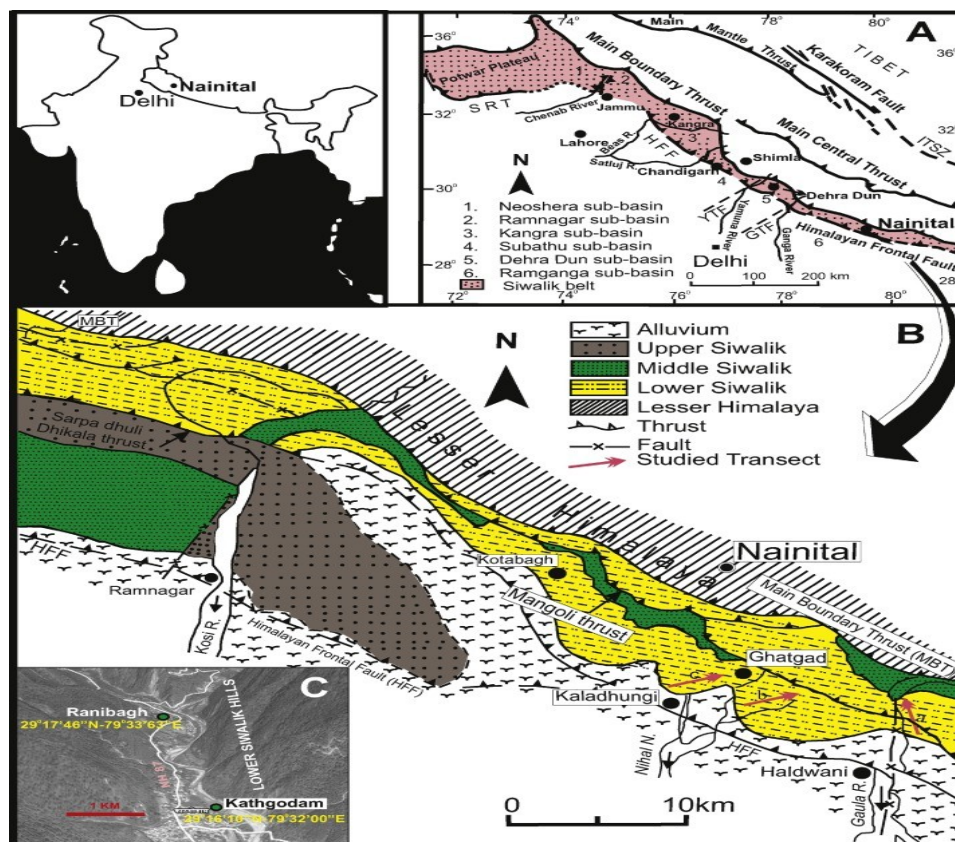


fig 1.1 : Siwalik Divisions.

1.1 Objective

- Locating ourselves on the toposheet with the help of GPS, or in case GPS is not present, then with the help of Brunton Compass, we need to take forward and backward reading (taking two reference points which are present on the toposheet).
- Mapping of lithology and preparation of geological map.
- Measurement of strike and dip with the help of Brunton Compass (for this, the primary need is the identification of bedding plane.)
- Recognition of primary sedimentary structure, i.e., ripple marks, trough, cross-bedding, etc.
- Recognition of structural features, i.e., folds, faults, joints, etc.
- Construction of geological history of that area, i.e., order of superposition.

CHAPTER 2. Study Area:

Our study area in the Mohand River basin, nestled within the Siwalik mountain range, holds immense geological significance. Situated within the Himalayan foreland basin, this region offers a unique opportunity to explore the dynamic interactions between the Himalayan mountain range and its surrounding foreland. The extensive alluvial fan formations present in the basin provide a natural laboratory for studying sediment transport, deposition, and landscape evolution. By examining sedimentary structures, grain size distributions, and lithological variations, we can reconstruct the paleoenvironmental history of the region. This basin's geological evolution is intricately linked to the tectonic upliftment associated with the growth of the Himalayas, allowing us to uncover the ancient geological history shaped by tectonic forces and climatic changes. Our expedition to the Mohand River basin serves as the focal point for our field report, enabling us to gather firsthand data and observations to contribute to a comprehensive analysis of sedimentary processes and their implications for understanding Earth's geological history.

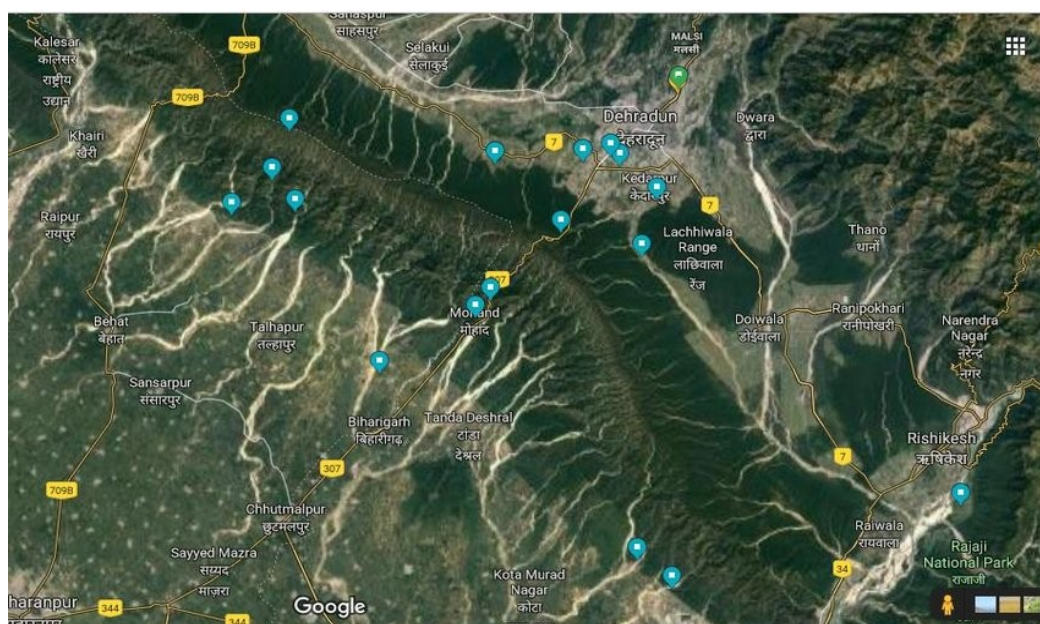


fig2.1 : Mohand River basin.

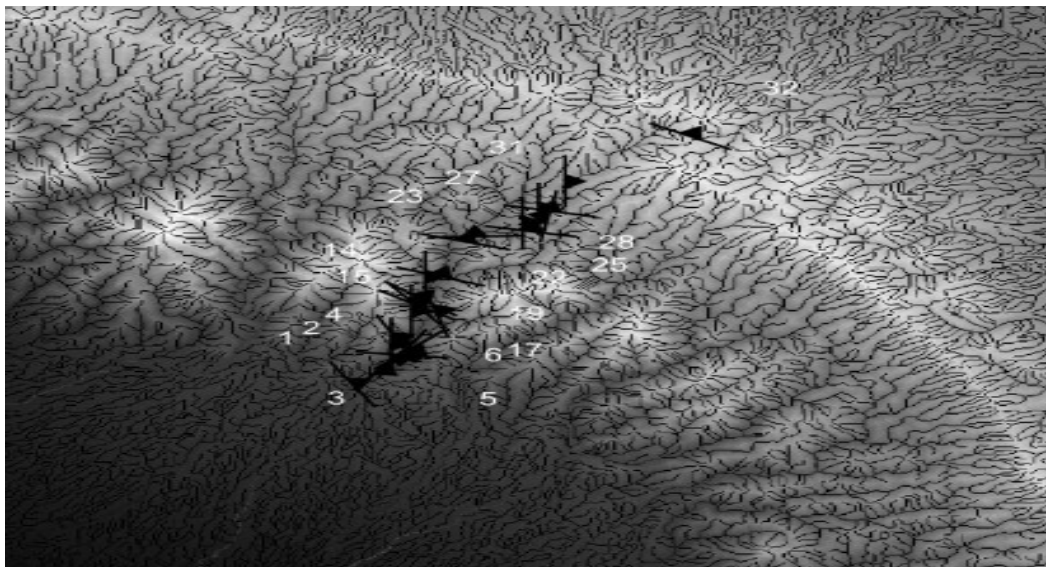


Fig 2.2 : Mohand River basin (GIS Plot)

2.1 Field Data Collection:

In this report, sedimentological and structural data are presented in separate sections. We begin with the structural data.

Structural Data: Strike and dip measure.

The structural data collection involved measurements of the dip, strike, and dip amount of bedding planes. To measure the dip, the bridge of the clinometer was positioned along the slope, and the value where the needle hung freely was noted. This value represented the dip amount of the bedding plane. Subsequently, a line was marked along the short length of the bridge. The bridge of the clinometer was then set perpendicular to this line, with the dial positioned to ensure that the needle remained parallel and not inclined in any direction. This procedure yielded the strike value. Adding 90° to the dip amount provided the dipping direction. At each location, a minimum of five readings was taken, and the values were averaged. The recorded values are presented as follows:



Fig 2.3 : Strike and dip measurement

Location	Longitude	Latitude	DMS(long)	DMS(Lat)	Measurement	Strike	Dip
	1E77°54'08.47"	N30°10'27.6"	77.9	30.17	1	275	27
					2	354	32
	2E77°54'25.38"	N30°10'36.44"	77.91	30.18	1	35	30
					2	30	30
	3E77°54'35.56"	N30°10'36.48.02"	77.91	30.18	1	75	50
					2	100	60
	4E77°54'38.73"	N30°10'49.70"	77.91	30.18	1	20	40
					2	25	33
	5E77°54'40.09"	N30°10'52.49"	77.91	30.18	1	40	27
					2	38	30
	6E77°54'41.76"	N30°10'52.51"	77.91	30.18	1	60	40
					2	55	25
	7E77°54'45"	N30°10'55"	77.91	30.19	1	30	45
					2	50	50
	E77°54'45"	N30°10'55"			3	28	45
	8E77°54'33.93"	N30°11'3.44"	77.91	30.19			
	9E77°54'44.45"	N30°11'12.94"			1	345	35
	E77°54'44.45"	N30°11'12.94"			2	335	20
	E77°54'44.45"	N30°11'12.94"			3	325	30
	E77°54'44.45"	N30°11'12.94"			4	345	30
10	E77°54'52.53"	N30°11'26.28"	77.91	30.19	1	330	30
	E77°54'53.75"	N30°11'26.56"	77.91	30.19	2	320	25
					3	310	35
					4	330	30
					5	310	32
11					1	305	30
12	E77°54'51.37"	N30°11'50.10"	77.91	30.2	1	330	30
	E77°54'51.37"	N30°11'50.10"			2	335	35
13	E77°54'51"	N30°11'49"	77.91	30.21(BEDS)		315	13
					2	282	19
	E77°54'51"	N30°11'49"			3	330	12.5
	E77°54'51"	N30°11'49"			4	280	25
	E77°54'51"	N30°11'49"			5	285	25
	E77°54'51"	N30°11'49"				330	18
	E77°54'51"	N30°11'49"			2	264	19
	E77°54'51"	N30°11'49"			3	280	18
	E77°54'51"	N30°11'49"			4	295	19
	E77°54'51"	N30°11'49"			5	285	19
	E77°54'51"	N30°11'49"				315	28
				1(Coal side beds)			
	E77°54'51"	N30°11'49"			2	298	29
	E77°54'51"	N30°11'49"				325	10
				1(Forsets)			

E77°54'51"	N30°11'49"			2	315	12
E77°54'51"	N30°11'49"			3	350	10
E77°54'51"	N30°11'49"			4	340	1
14E77°54'53"	N30°11'53"	77.91	30.2			
15E77°54'56"	N30°12'7"	77.91	30.2	1	280	30
16				1	335	28
17				1	330	33
				2	293	25
				3	315	21
18E77°55'1"	N30°12'12"	77.92	30.2	1	282	35
E77°55'1"	N30°12'12"			2	290	32
E77°55'1"	N30°12'12"			3	295	35
19E77°54'4"	N30°12'24"	77.9	30.21	1	305	31
E77°54'4"	N30°12'24"			2	300	35
E77°54'4"	N30°12'24"			3	275	30
E77°54'4"	N30°12'24"			4	290	32
20E77°55'77"	N30°12'45"	77.94	30.21	1	284	45
E77°55'77"	N30°12'45"			2	290	39
E77°55'77"	N30°12'45"			3	330	38
E77°55'77"	N30°12'45"			4	300	40
21E77°55'18"	N30°12'45"	77.92	30.21	1	284	45
				2	290	35
				3	290	39
				4	285	37
22E77°55'22"	N30°12'50"	77.93	30.221(Middle sibalik) Upper		285	30
					252	32
23E77°55'46"	N30°12'50"					
24E77°55'57.55"	N30°12'54.1 4"		Bed		305	25
E77°55'57.55"	N30°12'54.1 4"		forset		310	27
25E77°55'57.16"	N30°12'57.7 3"					
26E77°56'01.34"	N30°13'04.0 4"			1	285	2
27E77°56'12.84"	N30°13'06.1 3"					
28E77°56'10.06"	N30°13'33.3 5"				275	25

True thickness:

True thickness is perpendicular to the bedding or bounding plane
Apparent thickness: should be parallel to dip direct, else resolve it parallel to dip direction
 $b = a \sin\Theta$ (b: true thickness, a: apparent/outcrop thickness, Θ : dip/inclination angle)

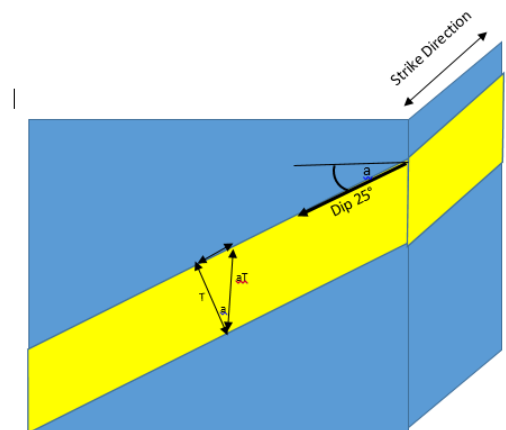


Fig 2.4 : True thickness

CHAPTER 3. Sedimentological Observations

In the sedimentary field, we conduct observations to study the formation and characteristics of sedimentary rocks. By analyzing sedimentary structures, fossils, grain sizes, and depositional environments, we gain insights into Earth's geological history. These observations help us understand past environmental conditions and the processes that have shaped our planet over time.

3.1 Identification of Sedimentary Rock,

In sedimentary rock identification, we assess key characteristics such as grain size, texture, composition, and sedimentary structures. By examining these features, we classify rocks into categories such as sandstone, shale, limestone, and conglomerate. This process allows us to infer the depositional environment and geological history of the rocks, providing valuable insights into Earth's past.

3.1.1 Primary Sedimentary Structures:

Here, we document and analyze primary sedimentary structures within the rock units. These structures provide valuable information about the depositional processes and conditions at the time of sediment deposition. Examples of primary sedimentary structures include bedding planes, cross-bedding, graded bedding, ripple marks, mud cracks, and bioturbation features such as burrows and tracks. By carefully documenting these structures, we can reconstruct the depositional environments and interpret the dynamics of sedimentary systems.



Fig 3.1: Primary features for identification of sedimentary rock

3.1.2 Grain Size Analysis:

Grain size analysis involves the collection and characterization of sediment samples from different locations within the field area. Using sieves and other analytical techniques, we separate sediment particles by size and determine the grain size distribution. This analysis provides insights into the transport and depositional processes that operated in the area. By examining the distribution of grain sizes, we can infer the energy of the depositional environment, the mode of sediment transport, and the sorting characteristics of the sedimentary deposit.



Fig 3.1: Grain size variation with 6 diff locations

3.1.3 Bedding Planes & Beds:

In this section, we focus on documenting the bedding planes and beds within the sedimentary rock units. Bedding planes represent the horizontal surfaces that separate individual layers or beds of sedimentary rock. Beds, on the other hand, are distinct layers of sedimentary rock that are separated by bedding planes. By analyzing the orientation, thickness, and lithology of bedding planes and beds, we can interpret depositional environments, depositional rates, and changes in sedimentary facies over time.



Fig 3.1: Bedding Planes & Bed at location5.

3.1.4 Lateral & Vertical Variations:

This section explores the lateral and vertical variations in sedimentary deposits within the field area. We document changes in lithology, sedimentary structures, and fossil content as we move laterally across the landscape or vertically through the stratigraphic sequence. By identifying these variations, we can reconstruct the spatial and temporal evolution of sedimentary environments and processes, providing insights into Earth's dynamic history.

CHAPTER 4. Interpretation of Sedimentary Environments

Dip & Strike Measurements using Stereonets and using clinometer & board:

To determine the strike of a geological feature using a clinometer, we align the long axis of the clinometer parallel to an imaginary horizontal line, as determined by the right hand rule (with the thumb pointing in the direction of strike and fingers indicating the dip direction). Tilt the clinometer until the needle moves freely, and the outermost reading on the clinometer indicates the strike value.

Once the horizontal line is established, we measure the dip value by orienting the long axis of the clinometer perpendicular to this horizontal line.

4.1: Lithologs :

Measurement of True thickness of a bed

For this, a certain section of the outcrop was chosen and the distance between the two end points was marked. The strike, dip and the dipping direction were measured at either of the two ends.

The results were then plotted on a graph sheet and with the help of the relation $b = a \sin \theta$ (b: true thickness, a: apparent/outcrop thickness, θ : dip/inclination angle), the true thickness of the bed was obtained.

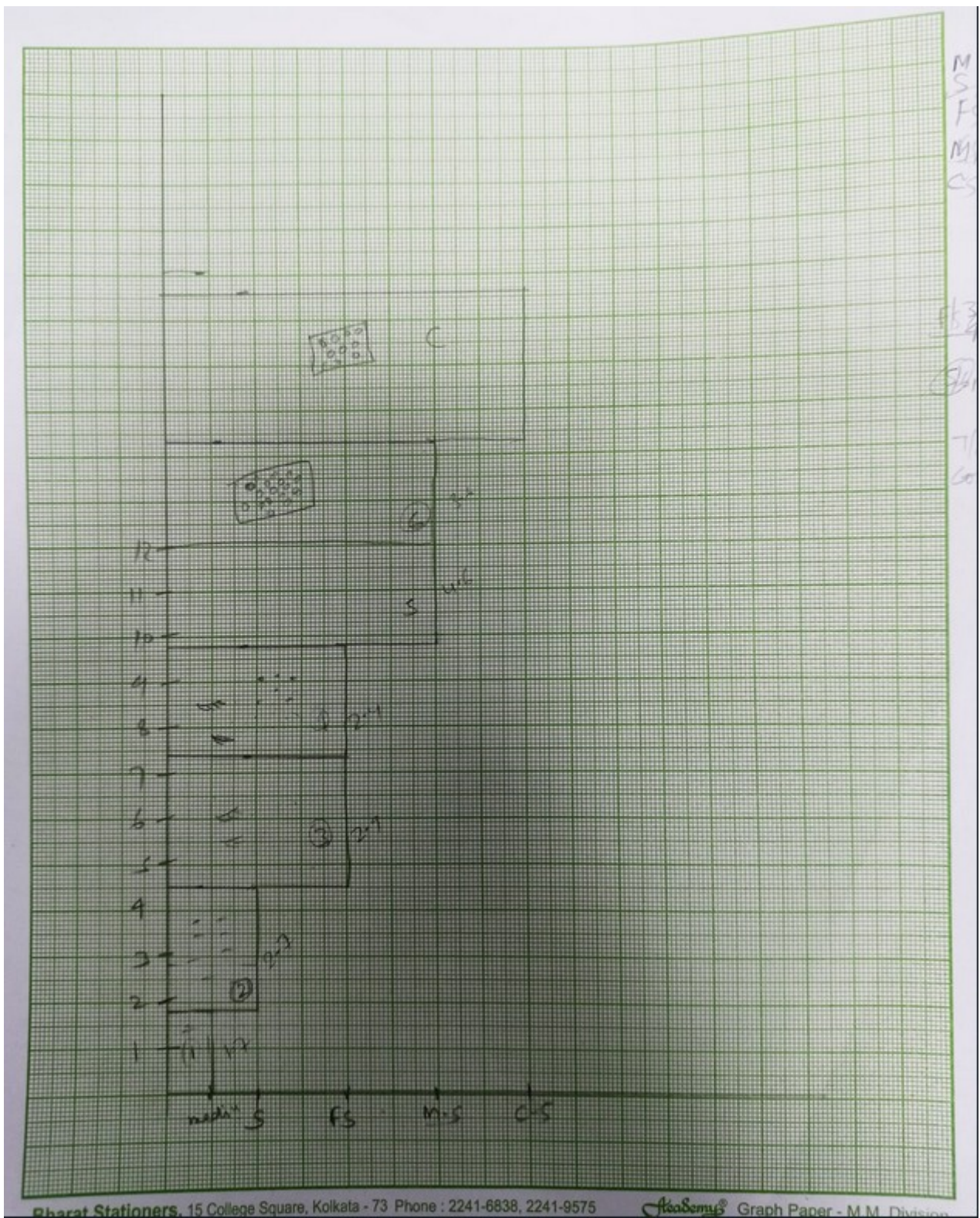


fig4.1 litholog

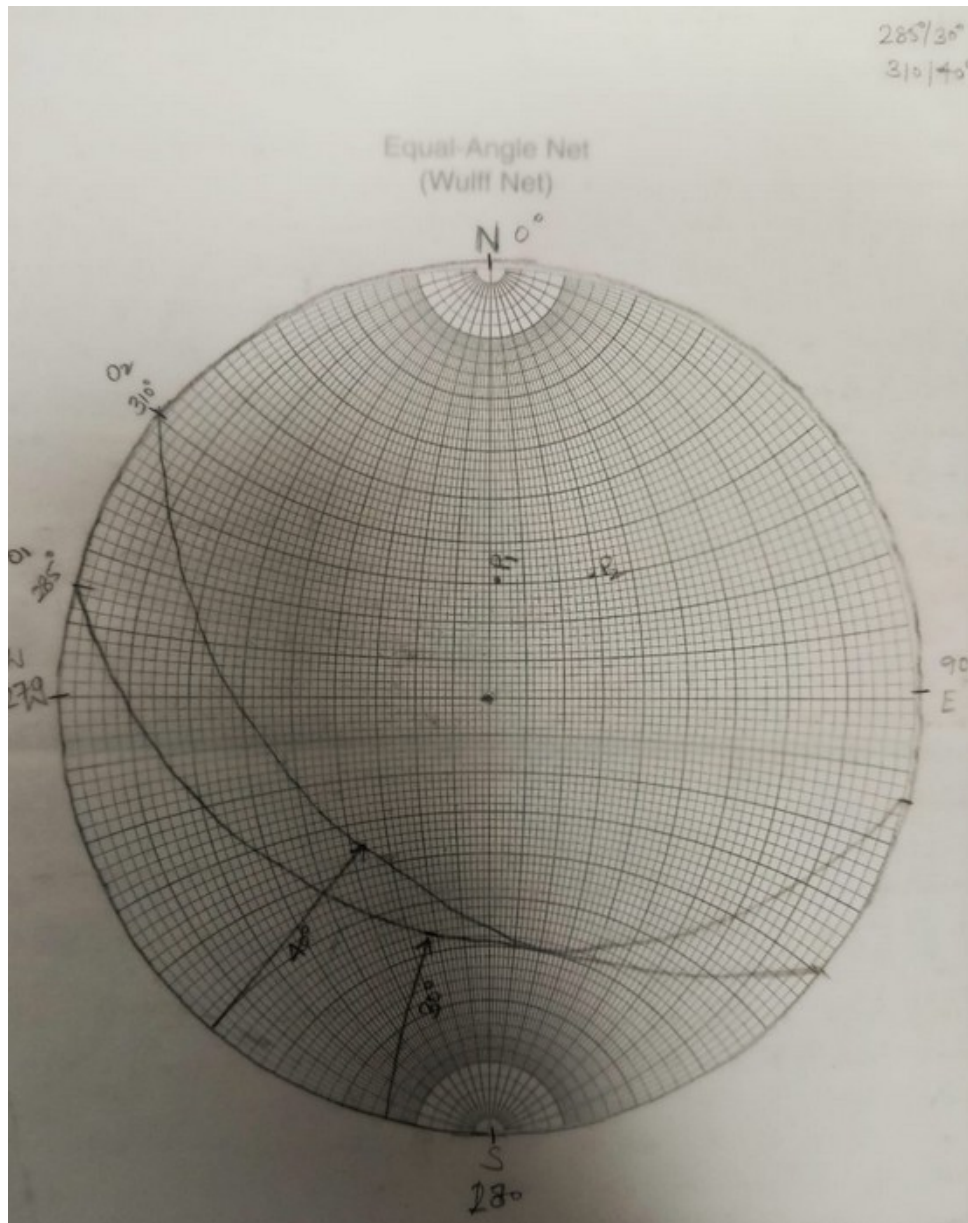


fig 4.2 stereonet plot

4.3: Geodynamic Interpretations-

Geodynamic interpretations in sedimentology involve understanding the geological processes and tectonic events that influence sedimentary deposition and the subsequent transformation of sedimentary rocks. One example of such interpretation is explaining the transition from fine sand to conglomerate within a sedimentary sequence.

The conversion from fine sand to conglomerate typically occurs due to changes in sediment supply, energy conditions, and tectonic activity over time. Here's a potential geodynamic interpretation for this transition:

CHAPTER 5- GEOCHEMICAL & PETROGRAPHIC STUDIES

Methodology

Powdering for XRF Data:

Powdering of sediment samples for XRF (X-ray fluorescence) analysis involves several steps to ensure accurate and representative results.

1. **Sample Preparation:** Start by selecting representative sediment samples from the field, ensuring they are free from contamination and weathering.
2. **Drying:** Dry the samples thoroughly to remove any moisture, as moisture content can affect XRF measurements.
3. **Grinding:** Once dried, grind the samples into a fine powder using a mortar and pestle or a mechanical grinder. The aim is to achieve a homogeneous and fine-grained powder to ensure uniform XRF analysis results.
4. **Homogenization:** After grinding, thoroughly mix the powder to ensure homogeneity and minimize any potential sampling bias.
5. **Pressing Pellets :** The powdered sample is pressed into pellets using a hydraulic press. Pelletization helps to achieve better analytical precision and reduces the risk of sample loss during analysis.
6. **XRF Analysis:**

Finally, analyze the powdered samples using X-ray fluorescence spectroscopy. XRF measures the elemental composition of the sample by bombarding it with X-rays, causing the atoms to emit characteristic fluorescent X-rays that are then detected and quantified. The resulting data provide information about the elemental composition of the sediment sample.



Fig .5.1: XRF machine



Fig; 5.2 powdering machine

Procedure of Making Thin Sections:

Thin sections are thin slices of rock or sediment that are mounted on glass slides and used for microscopic examination to study mineralogy, texture, and sedimentary structures. The procedure for making thin sections typically involves the following steps:

1. **Sample Selection:** Choose a representative sediment or rock sample that exhibits the features of interest.
2. **Trimming:** Trim the sample to a suitable size and shape using a rock saw or other cutting tools. The aim is to obtain a flat surface that will be polished to create the thin section.
3. **Mounting:** Glue the trimmed sample onto a glass slide using an adhesive such as epoxy resin. Ensure that the sample is securely attached to the slide to prevent movement during the cutting and polishing process.

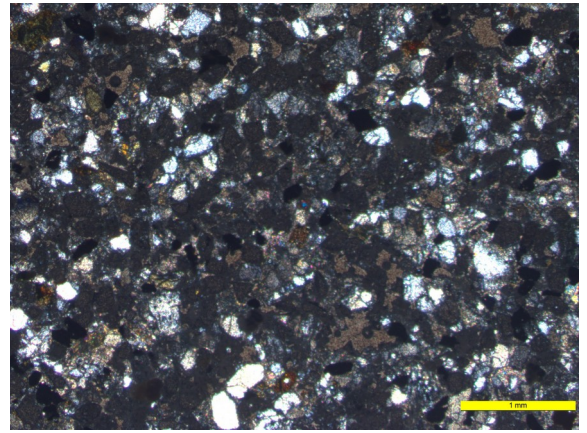
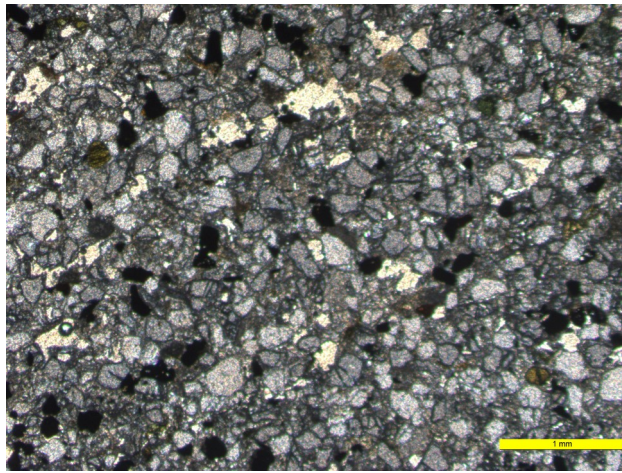


Fig 5.3 S-1 XPL, PPL

4. Grinding and Polishing: Utilize a lapidary grinder and polisher to gradually reduce the thickness of the sample to approximately 30-50 micrometers. This step involves the systematic removal of material from the sample's surface until it achieves the desired thinness.
5. Mounting on Glass Slide: Upon reaching the desired thinness, affix the sample onto a glass slide using a transparent mounting medium such as Canada balsam or epoxy resin. To safeguard the thin section and ensure optical clarity, cover it with a cover slip.
6. Final Polishing: Employ fine abrasive materials to polish the thin section, aiming for a smooth and optically clear surface suitable for microscopic examination.

XRF Result:

Sample Name	Measurement Finished	Calibration Method	Na2O (%)	MgO (%)	K2O (%)	CaO (%)	Fe2O3 (%)	Al2O3 (%)	SiO2 (%)	P2O5 (%)	MnO (%)	TiO2 (%)	ZrO2 (PPM)
JLK-1(07-02-24)	07/02/24 14:37	sediments-11-6-18	1.09	1.62	2.83	0.68	7.36	16.05	58.55	0.1902	0.27	0.652	187
JSD-2(07-02-24)	07/02/24 14:20	sediments-11-6-18	2.45	2.85	1.21	3.72	11.6	13.78	61.18	0.1294	0.11	0.6091	150
SAMPLE -3 Gr-4 RPT	07/02/24 16:51	sediments-11-6-18	1.92	0.84	1.06	0.23	2.62	10.26	89.56	0.0853	0.03	0.437	276.3
JLK-1(07-02-24)	07/02/24 16:43	sediments-11-6-18	1.09	1.62	2.83	0.68	7.35	16.06	58.53	0.1896	0.27	0.6525	187.3
SAMPLE -3 Gr-3 RPT	07/02/24 16:35	sediments-11-6-18	1.68	1.17	1.58	8.27	1.27	11.11	63.43	0.0939	0.07	0.5963	156.8
JSD-2(07-02-24)	07/02/24 16:26	sediments-11-6-18	2.45	2.85	1.21	3.72	11.59	13.78	61.17	0.1302	0.11	0.6094	150.4
SAMPLE -9 Gr-3	07/02/24 16:18	sediments-11-6-18	2.46	1.46	1.76	2.28	2.53	11.49	70.91	0.0961	0.07	0.5488	239.6
SAMPLE -3 Gr-4	07/02/24 16:10	sediments-11-6-18	1.91	0.82	1.06	0.23	2.6	10.22	88.89	0.0852	0.03	0.4411	276.8
SAMPLE -8 Gr-3	07/02/24 16:01	sediments-11-6-18	1.27	0.94	1.16	1.12	11.99	11.92	57.2	0.11	0.26	0.6852	395
SAMPLE -5 Gr-4	07/02/24 15:53	sediments-11-6-18	2.03	1.47	2.19	1.36	3.18	12.27	60.2	0.107	0.05	0.6286	360.1
SAMPLE -4 Gr-4	07/02/24 15:44	sediments-11-6-18	1.52	1.06	1.05	7.97	7.07	10.34	62.21	0.1382	0.27	0.6093	1244.4
SAMPLE -2 Gr-4	07/02/24 15:36	sediments-11-6-18	1.53	1.23	1.69	0.41	4.43	11.81	69.71	0.1193	0.06	0.5454	227.7
SAMPLE -7 Gr-3	07/02/24 15:28	sediments-11-6-18	2.2	1.41	2.34	0.75	2.39	12.31	68.53	0.0922	0.04	0.5652	255.9
SAMPLE -6 Gr-3	07/02/24 15:19	sediments-11-6-18	1.74	1.02	1.17	0.32	2.68	10.6	77.1	0.1065	0.04	0.5027	351.1
SAMPLE -5 Gr-3	07/02/24 15:11	sediments-11-6-18	1.57	1.21	1.54	0.35	2.82	11.43	74.41	0.097	0.06	0.5115	266.5
SAMPLE -3 Gr-3	07/02/24 15:02	sediments-11-6-18	1.68	1.16	1.58	8.26	1.26	11.07	63.24	0.0941	0.07	0.5971	156.8
SAMPLE -2 Gr-3	07/02/24 14:53	sediments-11-6-18	0.85	0.14	0.35	0.69	1.81	7.57	93.63	0.5935	0.02	0.5659	88.7
SAMPLE -1 Gr-3	07/02/24 14:45	sediments-11-6-18	1.55	1.11	1.06	8.32	6.08	10.53	63.9	0.1453	0.27	0.5842	671.3

Table 2

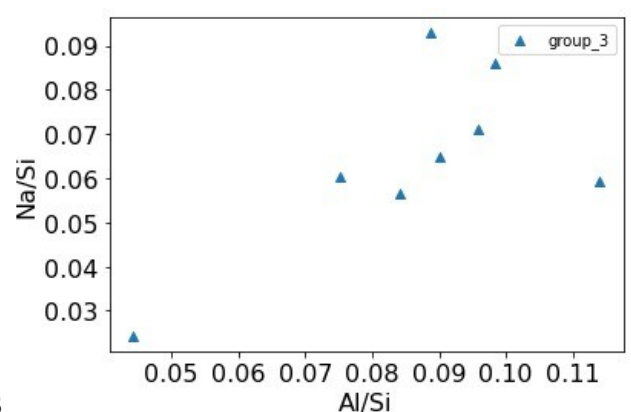
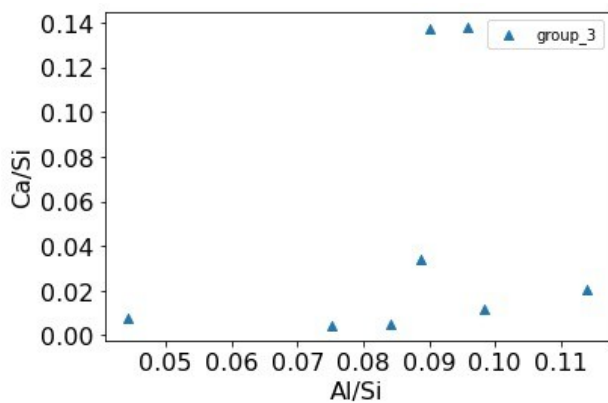


fig 5.3

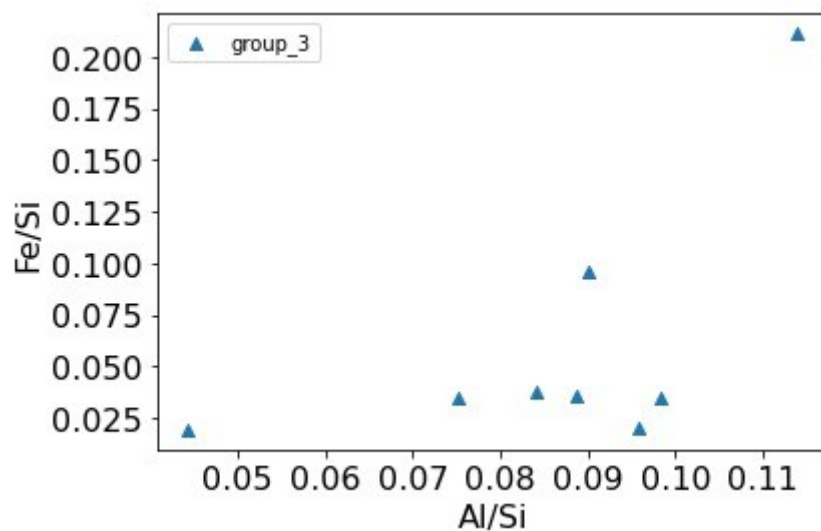


Fig 5.4

CHAPTER 6 -TECTONICS FROM SEDIMENTARY RECORDS

Here titled "Tectonics from Sedimentary Records," the Mohand Range in the Himalayan Foreland Basin serves as a remarkable case study for understanding regional tectonic evolution. The sedimentary sequences observed therein offer valuable insights into the complex interaction between tectonic forces and depositional environments.

Beginning with the observation of meandering channels in distal areas and transitioning to braided channels upstream, and eventually to alluvial deposits in proximal regions, reveals a dynamic relationship between tectonics and sedimentation. The presence of meandering channels in distal regions implies relatively calm depositional conditions, possibly reflecting stable tectonic settings conducive to the formation of sinuous river courses.

As one progresses upstream, the appearance of braided channels indicates heightened tectonic activity and sediment input, leading to more energetic conditions and the fragmentation of river channels into multiple shallow streams. This shift towards braided channels signals tectonic uplift and increased erosion, particularly near the Himalayan uplift zone.

Additionally, the change in hill dip direction from southwest to northeast, along with the gradual shift in grain size from fine to coarse from distal to proximal locations, provides further evidence of tectonic influence. The northeastward dip of hills suggests a compressional tectonic regime, aligning with the northward movement of the Indian plate colliding with the Eurasian plate.

Furthermore, subtle yet significant geological features such as the directional change in hill dip from southwest to northeast provide additional clues about the prevailing tectonic regime. The northeastward dip of hills implies a compressional tectonic setting, which aligns with the broader context of the northward convergence of the Indian plate against the Eurasian plate.

Moreover, the gradational shift in grain size from fine to coarse sediment from distal to proximal locations corroborates the influence of tectonic forces on sediment transport and deposition. This gradual change in grain size reflects variations in sediment supply and transportation dynamics along the tectonically active landscape.

In essence, the sedimentary records within the Mohand Range serve as a dynamic archive, offering invaluable insights into the intricate interplay between tectonic processes and sedimentary

the sedimentary records within the Mohand Range offer valuable clues about the tectonic processes shaping the region, highlighting the intricate relationship between tectonics and sedimentary environments.

CHAPTER 7- SUMMARY

The sedimentological expedition conducted in the Mohand Rao sector of the Siwalik Himalayas. This expedition yielded profound insights into the geological dynamics of the region through meticulous data collection and thorough analysis of sedimentological, mineralogical, and geochemical characteristics. By scrutinizing sediment transport patterns, paleoenvironments, and tectonic influences, the expedition shed light on the complex interplay of geological processes.

The Mohand River basin, nestled within the Siwalik mountain range, provided a unique setting to study sediment transport, deposition, and landscape evolution. Structural data collection involved meticulous measurements of dip and strike, aiding in the interpretation of geological features. Sedimentological observations focused on identifying rock types, analyzing grain sizes, and documenting bedding planes and beds. These observations provided insights into past environmental conditions and depositional processes.

Interpretation of sedimentary environments involved geodynamic analyses, linking sediment deposition to tectonic events and landscape evolution. Geochemical and petrographic studies, including XRF analysis and thin section preparation, added depth to our understanding of sedimentary processes. Notably, tectonics from sedimentary records revealed the dynamic relationship between tectonic forces and depositional environments.

Through the identification of primary sedimentary structures and the interpretation of lateral and vertical variations, the expedition uncovered evidence of dynamic tectonic uplift, sediment deposition, and erosion processes. Particularly noteworthy was the transition observed from meandering to braided channels and eventually to alluvial deposits, which underscored the significant impact of ongoing tectonic activity on the landscape and sedimentary records of the Siwalik range.

The summary underscores the profound significance of fieldwork in unraveling the geological history of mountainous regions, emphasizing the intricate relationship between tectonics and sedimentology. By combining meticulous field observations with detailed analytical techniques, the expedition provided a holistic understanding of the geological dynamics at play in the Mohand Rao sector, contributing valuable insights to the broader understanding of mountain-building processes and landscape evolution.

*****THE END*****